MATERI PERKULIAHAN

ANATOMI & FISIOLOGI SISTEM HEMATOLOGI

Dari sejarah, pembentukan sel darah, hingga penerapan klinis.

Disusun oleh

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INSTITUT KESEHATAN RAJAWALI

Sejarah: dari sirkulasi hingga sel darah

Landasan ilmu hematologi dibangun lewat observasi anatomi dan kemajuan mikroskop.

1628

William Harvey

Menjelaskan sirkulasi darah yang dipompa jantung.

1674

A. van Leeuwenhoek

Mendeskripsikan eritrosit melalui mikroskop.

1868

Neumann & Bizzozero

Menghubungkan sumsum tulang dengan pembentukan sel darah.

1901

Karl Landsteiner

Mengidentifikasi kelompok golongan darah ABO.

Hematologi memasuki era modern

Dari pengamatan visual menuju diagnosis dan terapi berbasis sel.

1

MIKROSKOPI

Identifikasi bentuk eritrosit, leukosit dan trombosit.

2

LABORATORIUM

Hitung darah lengkap, apusan darah, dan pemeriksaan koagulasi.

3

KEDOKTERAN MODERN

Terapi penyakit darah, transfusi aman, dan transplantasi sel punca.

Mazzarello, 2020; Keohane dkk., 2024

Tujuan pembelajaran

Setelah materi ini, mahasiswa mampu:

01

MENJELASKAN

Sejarah, fungsi, dan komposisi darah.

02

MENGIDENTIFIKASI

Organ pembentuk darah serta karakteristik sel.

03

MENGHUBUNGKAN

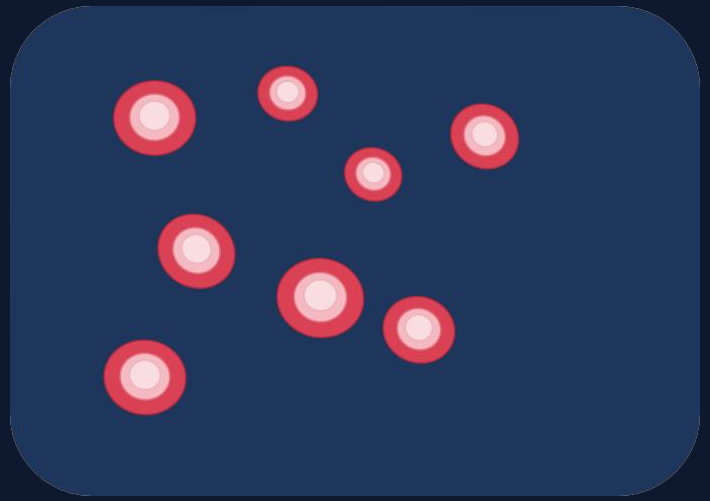
Proses fisiologi dengan pemeriksaan klinis.

Apa itu sistem hematologi?

Sistem yang mencakup darah, pembentukan sel darah, dan organ yang menunjang fungsinya.

DARAH

Medium transport dan perlindungan



LIMPA & NODUS LIMFA

Penyaringan dan fungsi imun

SUMSUM TULANG

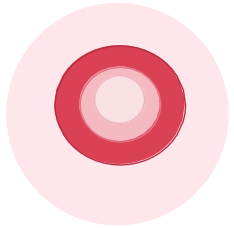
Pabrik utama sel darah

GINJAL & HATI

Pengaturan faktor pertumbuhan

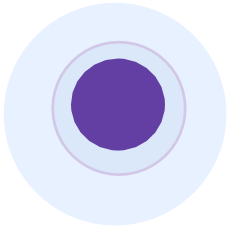
Betts dkk., 2022; Hoffbrand dkk., 2024

Tiga fungsi utama darah



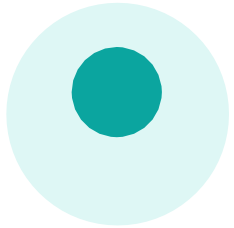
TRANSPORTASI

Oksigen, karbon dioksida, nutrisi, dan hormon.



PERLINDUNGAN

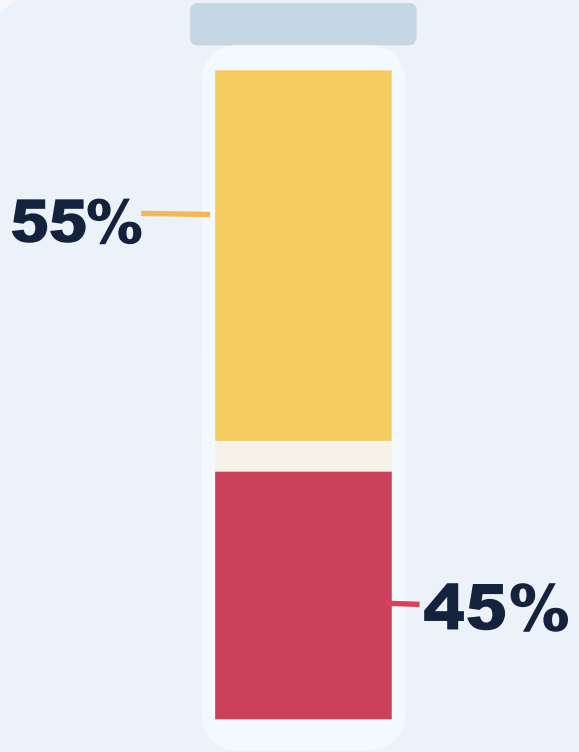
Leukosit melawan infeksi; trombosit membantu menghentikan perdarahan.



REGULASI

Menjaga pH, suhu tubuh, dan keseimbangan cairan.

Apa saja komponen darah?



PLASMA ±55%

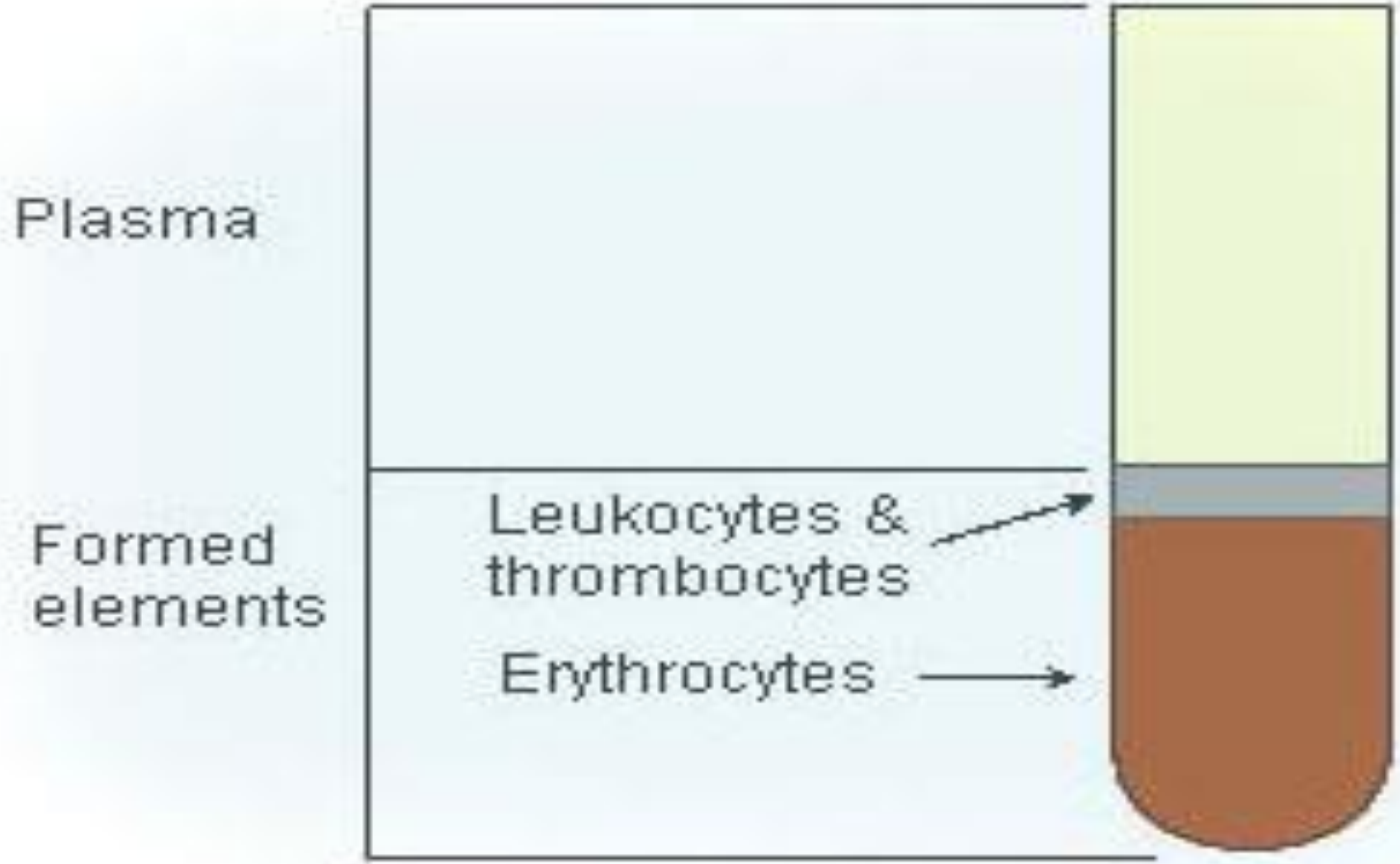
Air, protein dan zat terlarut

ERITROSIT ±45%

Bagian terbesar dari unsur berbentuk

BUFFY COAT <1%

Leukosit dan trombosit


Plasma
Formed elements
Leukocytes & thrombocytes
Erythrocytes

Plasma darah: lebih dari sekadar air

SEKITAR 90% AIR

Plasma menjadi medium pengangkut dan menjaga tekanan osmotik.

ALBUMIN

Mempertahankan tekanan onkotik dan mengangkut molekul.

GLOBULIN

Terlibat dalam sistem imun dan transport protein.

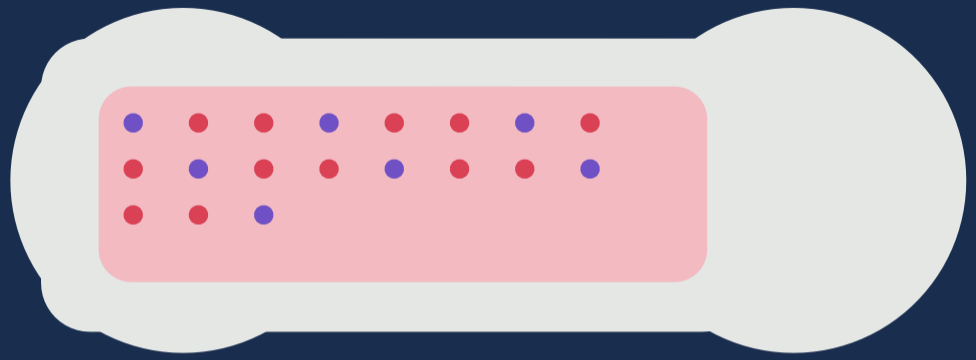
FIBRINOGEN

Protein penting dalam proses pembekuan darah.

Betts dkk., 2022

Sumsum tulang: lokasi produksi darah

Pada orang dewasa, hematopoiesis terutama terjadi di sumsum tulang merah.



SUMSUM TULANG MERAH

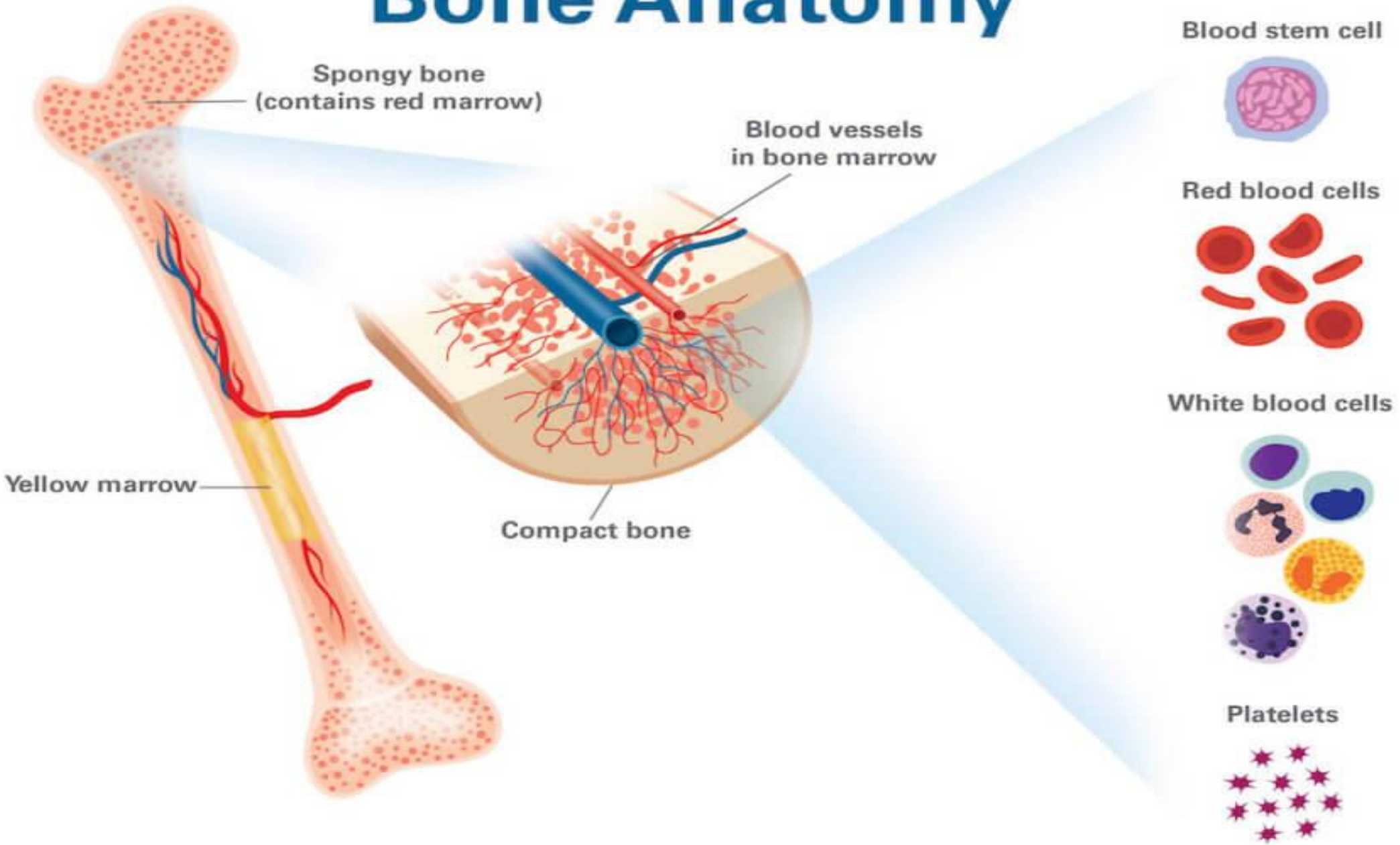
STERNUM
Tulang dada

VERTEBRA
Tulang belakang

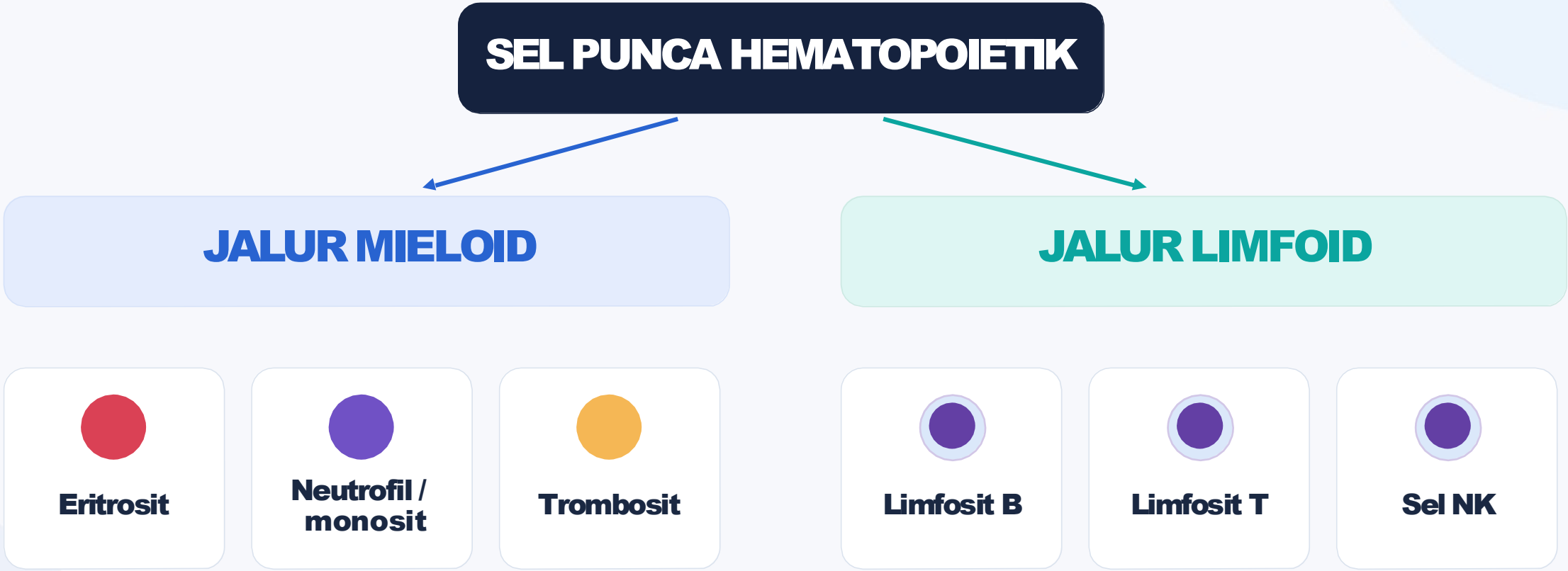
PELVIS
Tulang panggul

Betts dkk., 2022

Bone Anatomy

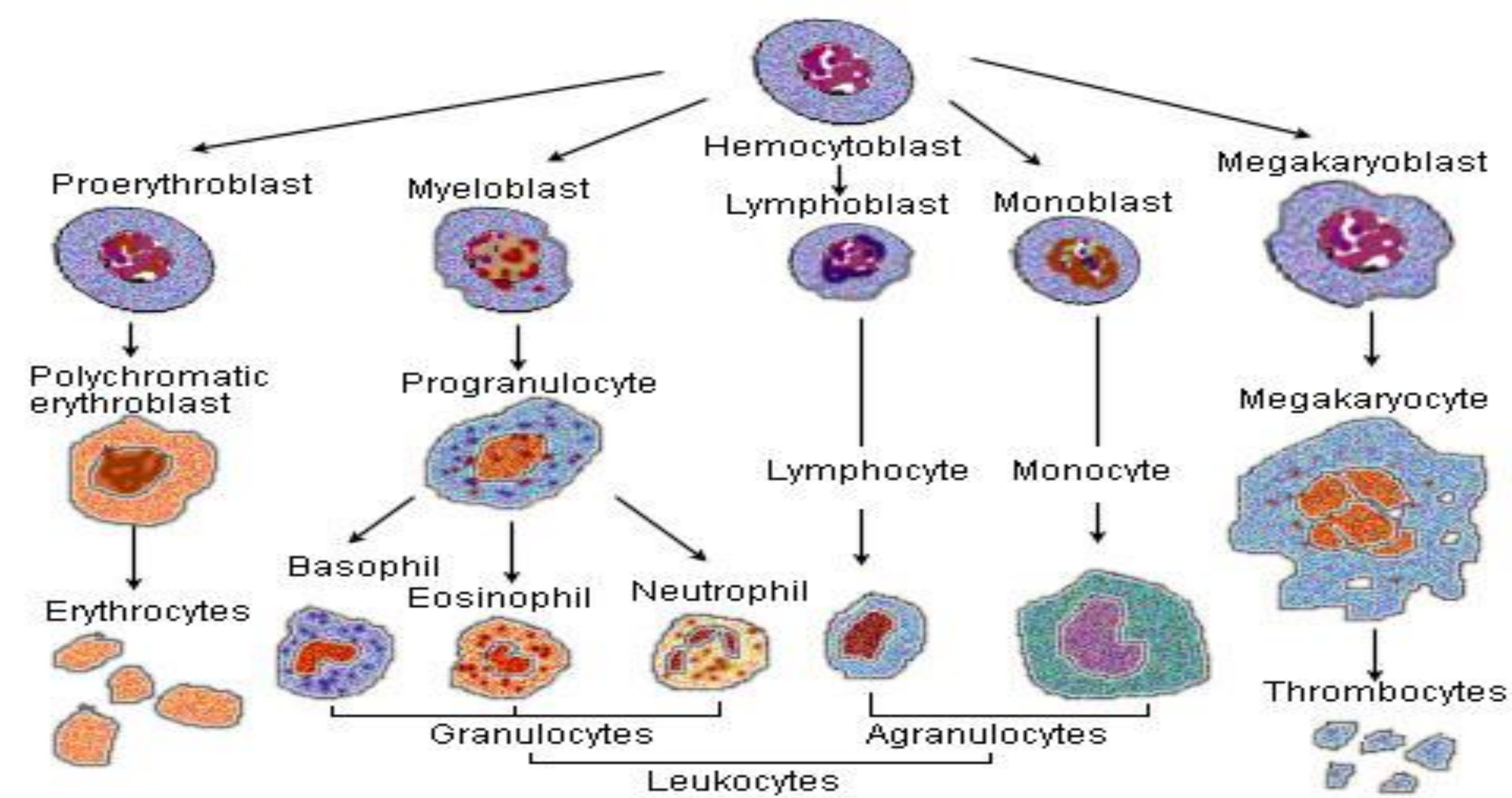


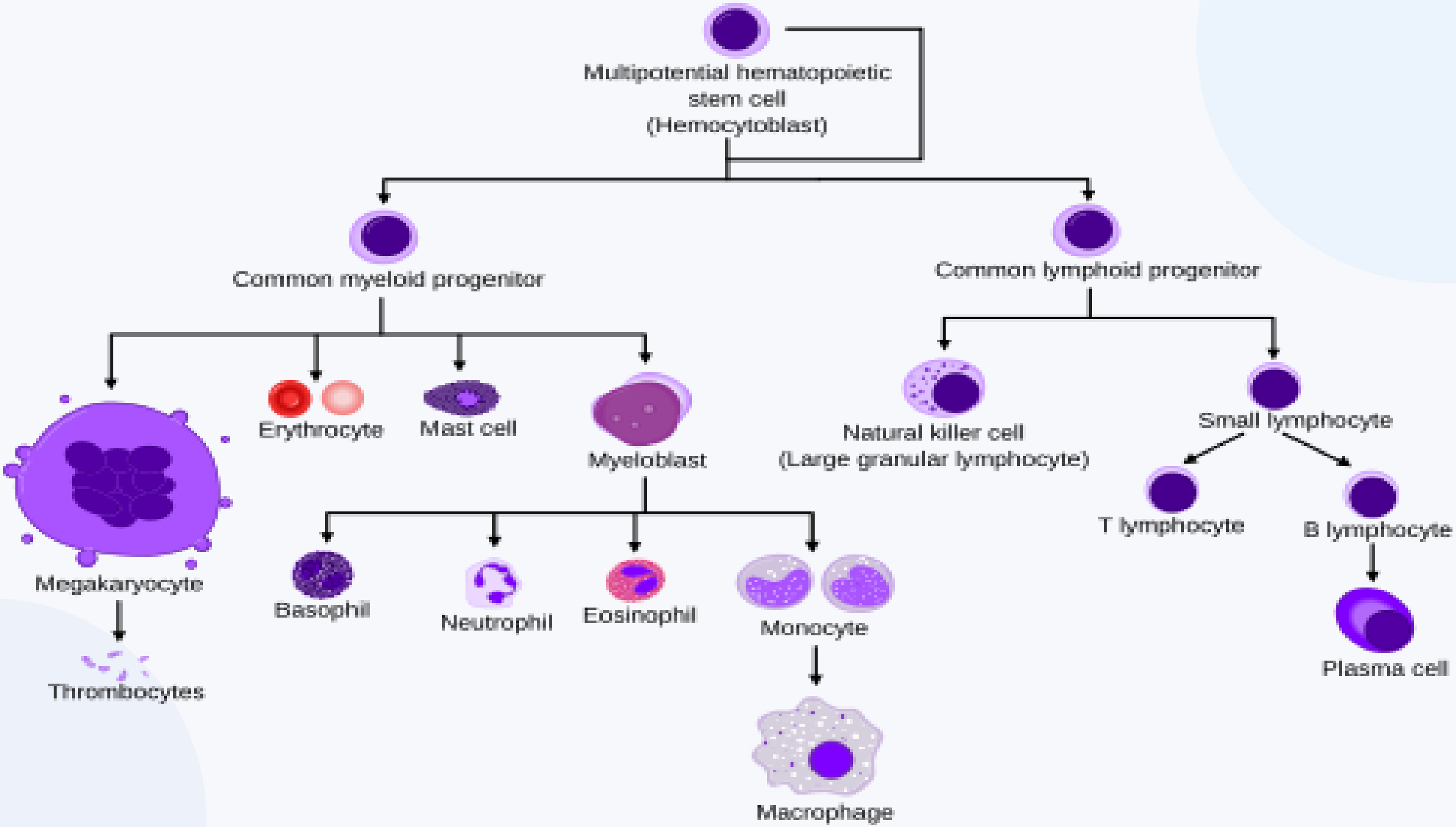
Hematopoiesis: asal semua sel darah



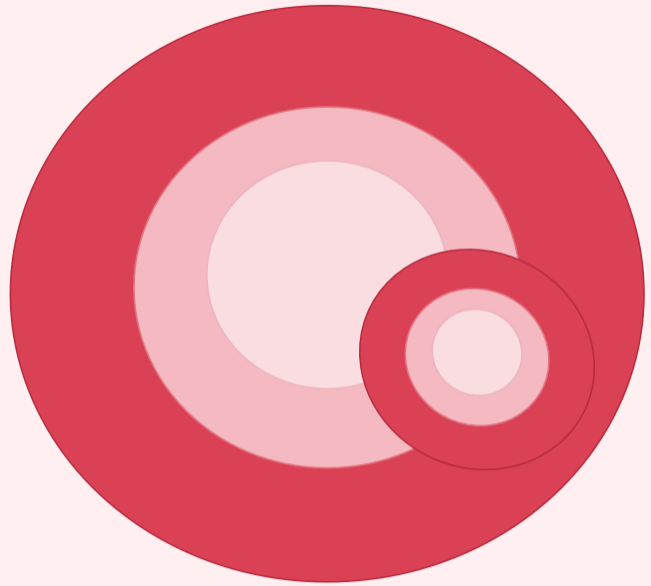
Sel punca → progenitor → sel matang

Betts dkk., 2022; Keohane dkk., 2024


Hemocytoblast
Proerythroblast
Myeloblast
Lymphoblast
Monoblast
Megakaryoblast
Polychromatic erythroblast
Progranulocyte
Megakaryocyte
Lymphocyte
Monocyte
Basophil
Eosinophil
Neutrophil
Erythrocytes
Thrombocytes
Granulocytes
Agranulocytes
Leukocytes


Multipotential hematopoietic stem cell (Hemocytoblast)
Common myeloid progenitor
Common lymphoid progenitor
Erythrocyte
Mast cell
Myeloblast
Natural killer cell (Large granular lymphocyte)
Small lymphocyte
Megakaryocyte
Basophil
Neutrophil
Eosinophil
Monocyte
T lymphocyte
B lymphocyte
Thrombocytes
Plasma cell
Macrophage

Eritrosit: pembawa oksigen



BENTUK BIKONKAF

7–8 µm

diameter rerata

±120 hari

masa hidup

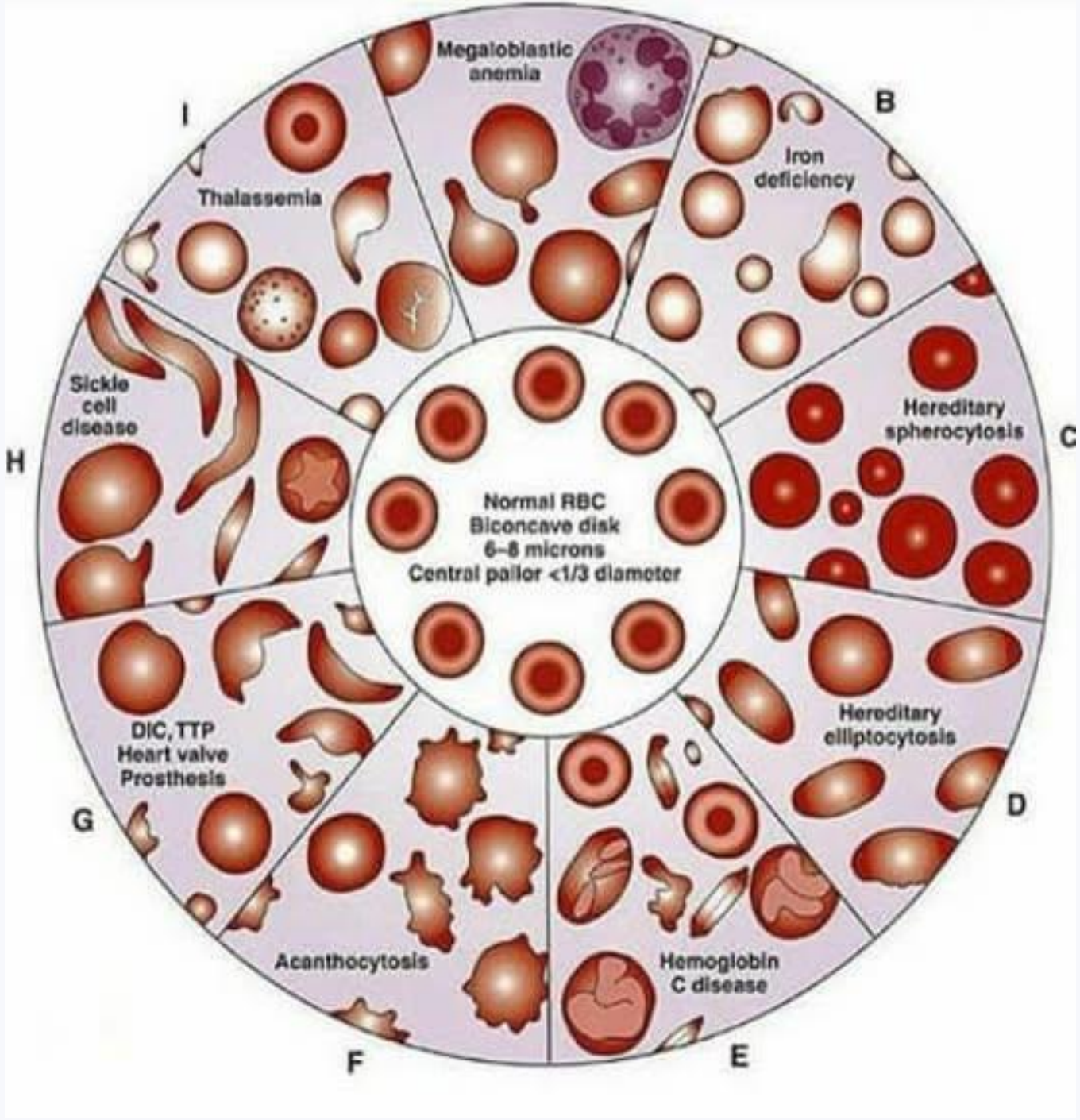
Tanpa inti

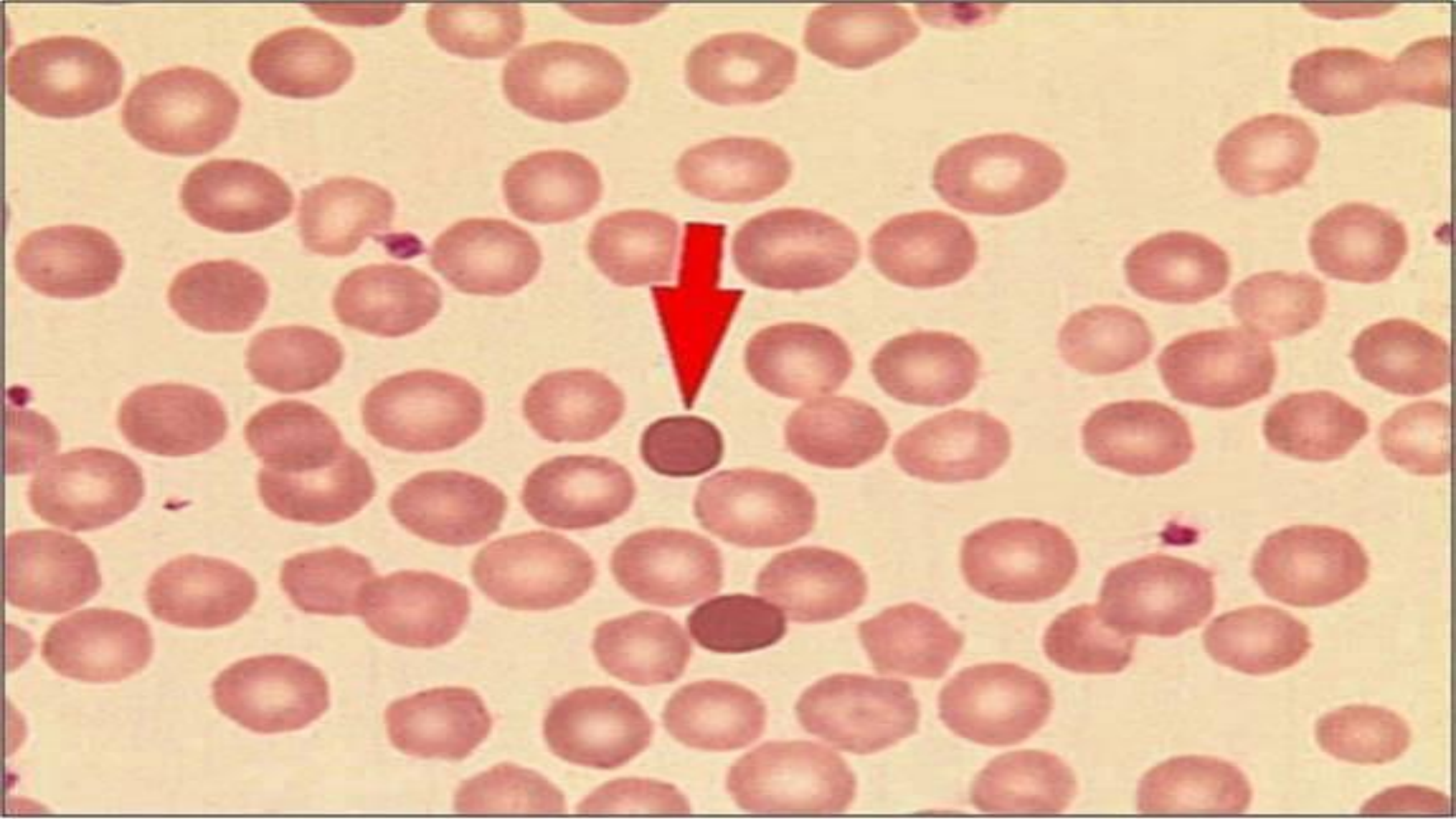
eritrosit matang

Hemoglobin

mengikat oksigen

Betts dkk., 2022; Hoffbrand dkk., 2024


Megaloblastic anemia
I
Thalassemia
B
Iron deficiency
Sickle cell disease
H
Normal RBC
Biconcave disk
6–8 microns
Central pallor <1/3 diameter
Hereditary spherocytosis
C
DIC, TTP
Heart valve
Prosthesis
G
Hereditary elliptocytosis
D
Acanthocytosis
F
Hemoglobin C disease
E



Hemoglobin mengangkut oksigen



PARU-PARU

Oksigen masuk ke darah



ERITROSIT

Hb mengikat dan membawa O_2



JARINGAN

Oksigen dilepas untuk sel

KUNCINYA: HEME + BESI

Hall & Hall, 2025; Betts dkk., 2022

Siklus hidup eritrosit

01

GINJAL

Hipoksia merangsang produksi EPO.

02

SUMSUM TULANG

EPO memicu pembentukan eritrosit.

03

SIRKULASI

Eritrosit bekerja sekitar 120 hari.

04

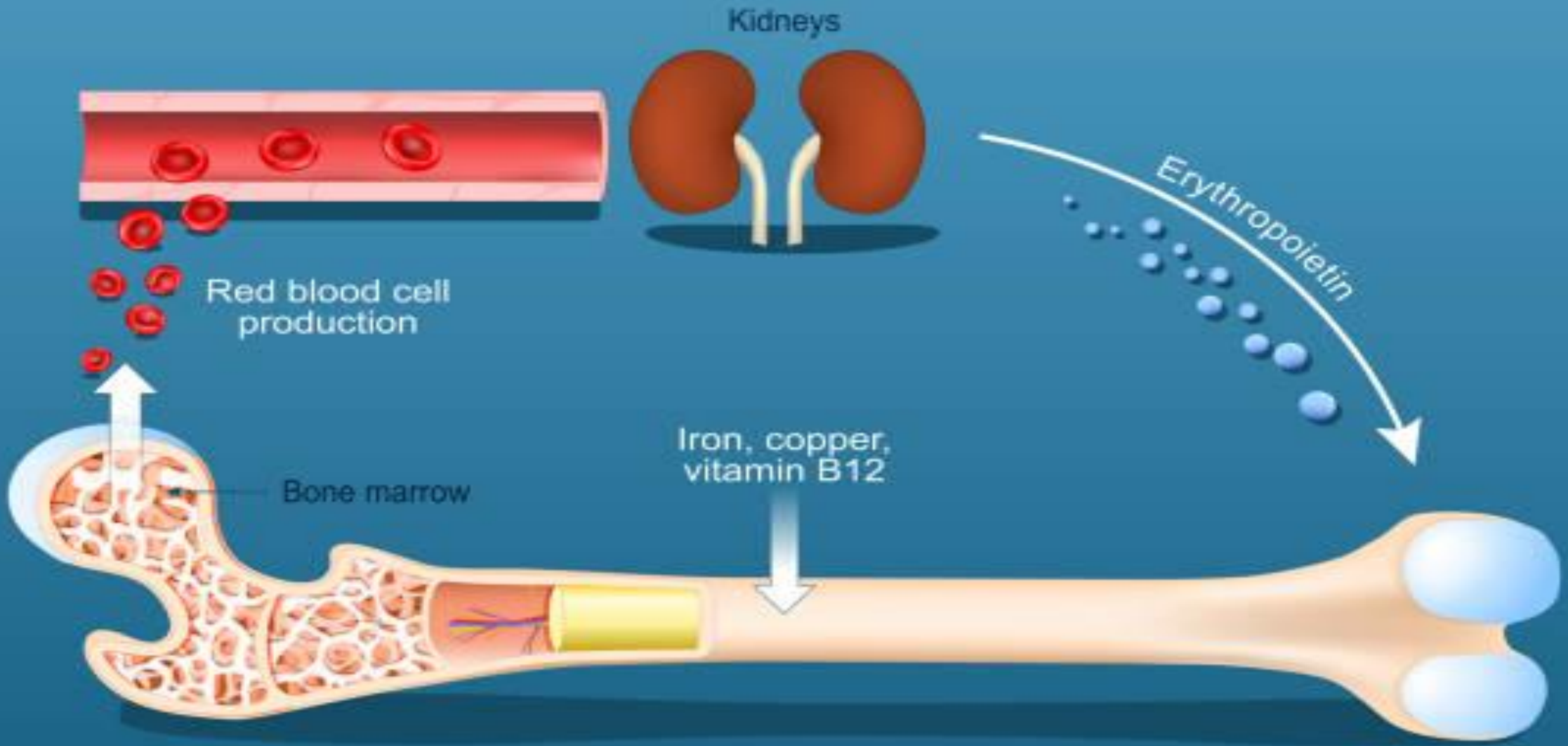
LIMPA & HATI

Eritrosit tua diurai; besi didaur ulang.

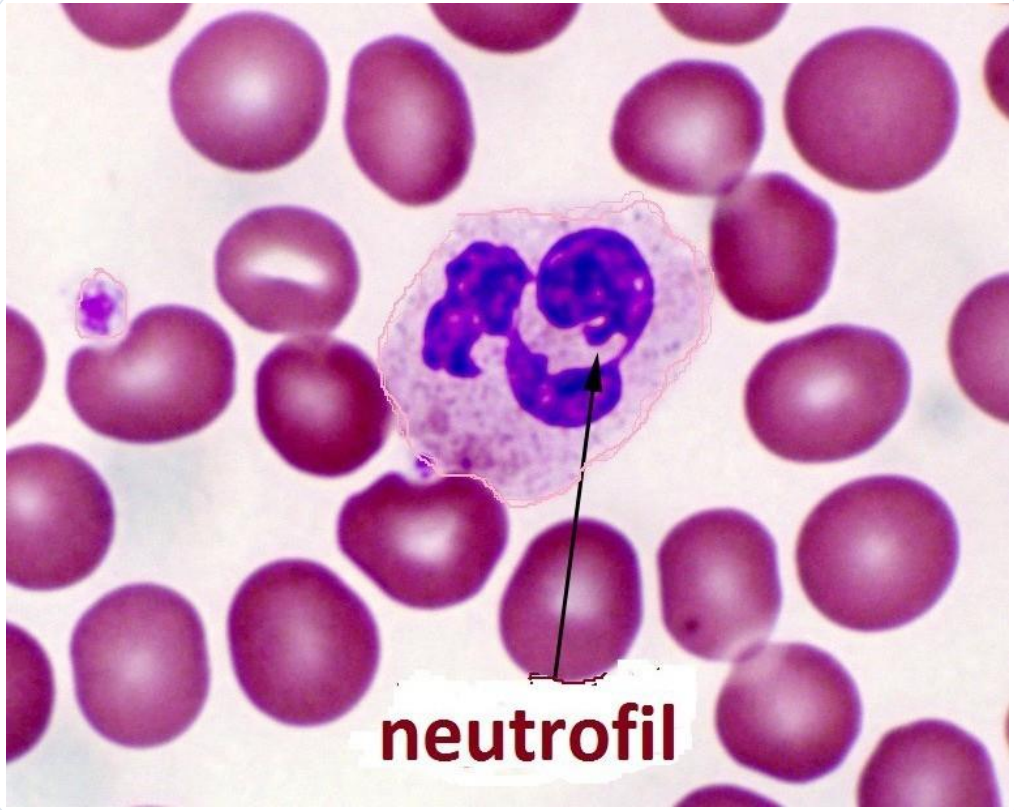
EPO = eritropoietin

Betts dkk., 2022; Hall & Hall, 2025

Erythropoietin



Leukosit: penjaga pertahanan tubuh



FAGOSITOSIS

Neutrofil dan monosit menelan patogen.

IMUNITAS ADAPTIF

Limfosit B dan T mengenali target spesifik.

RESPONS INFLAMASI

Mediator sel imun membantu respons terhadap ancaman.

Betts dkk., 2022; gambar dari materi pengguna

Types of Leukocytes (White Blood Cells)

The Biology Notes | The Chemistry Notes | Created with bio RENDER



Neutrophil



Eosinophil



Basophil



Monocyte



Dendritic cell



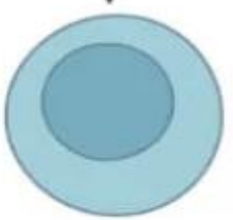
Macrophage



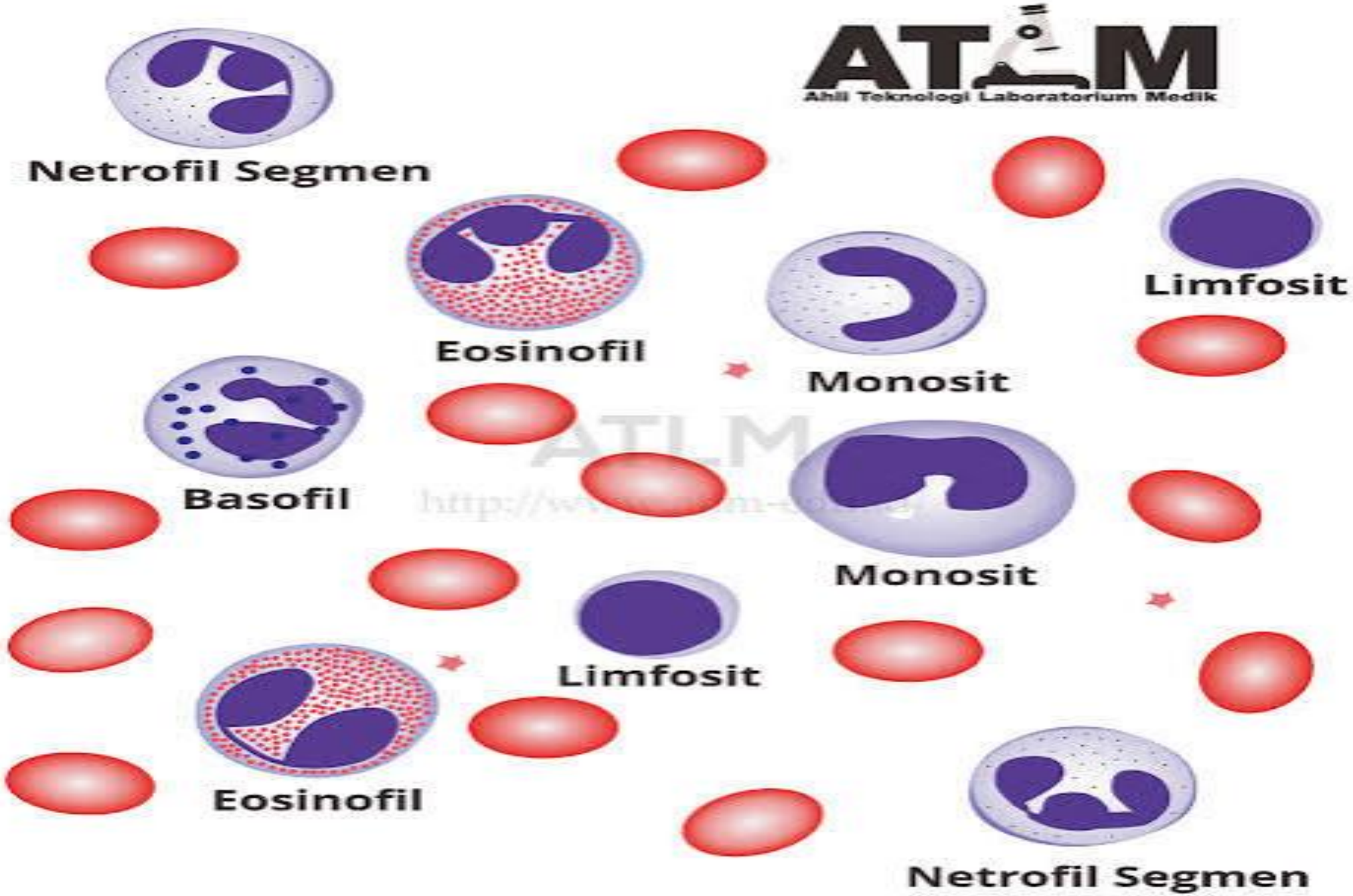
Lymphocyte



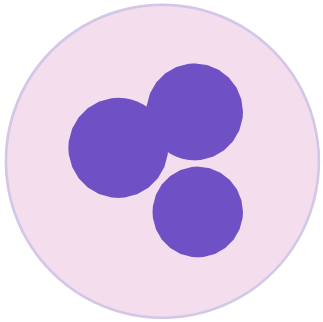
T cell



B cell

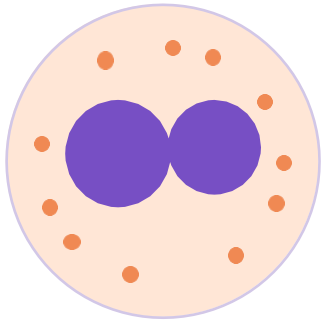

ATLM
Ahli Teknologi Laboratorium Medik
Netrofil Segmen
Eosinofil
Limfosit
Monosit
Basofil
Monosit
Limfosit
Eosinofil
Netrofil Segmen

Granulosit dan fungsi utamanya



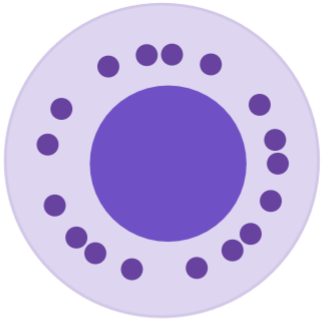
NEUTROFIL

Pertahanan awal terhadap bakteri.



EOSINOFIL

Berperan pada infeksi parasit dan alergi.

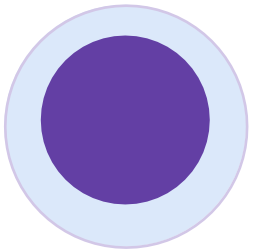


BASOFIL

Melepas mediator seperti histamin.

Betts dkk., 2022; Keohane dkk., 2024

Agranulosit: limfosit dan monosit

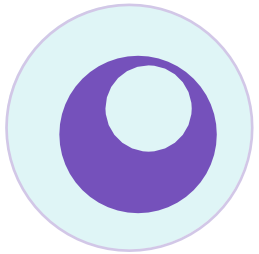


LIMFOSIT

Sel B → membuat antibodi

Sel T → mengatur / membunuh sel terinfeksi

Sel NK → membunuh sel abnormal



MONOSIT

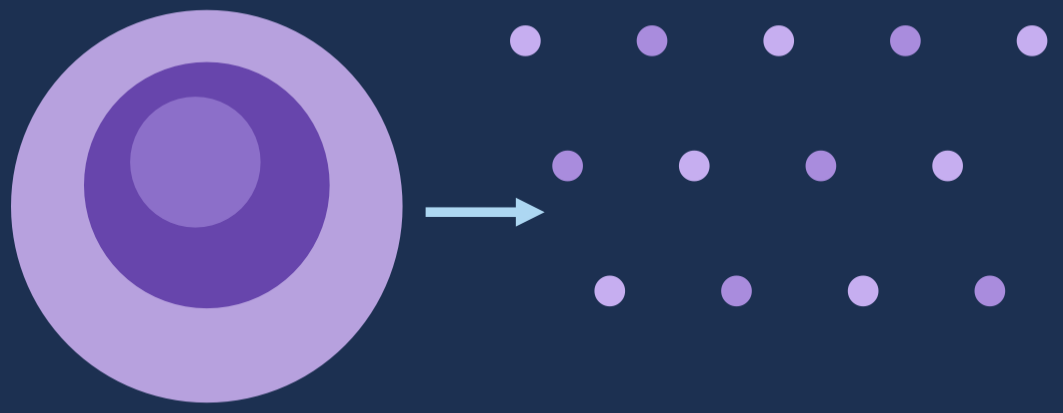
Masuk ke jaringan dan dapat berkembang menjadi makrofag.

FAGOSITOSIS

Betts dkk., 2022; Hoffbrand dkk., 2024

Trombosit: komponen awal hemostasis

Trombosit adalah fragmen sel kecil tanpa inti, berasal dari megakariosit.

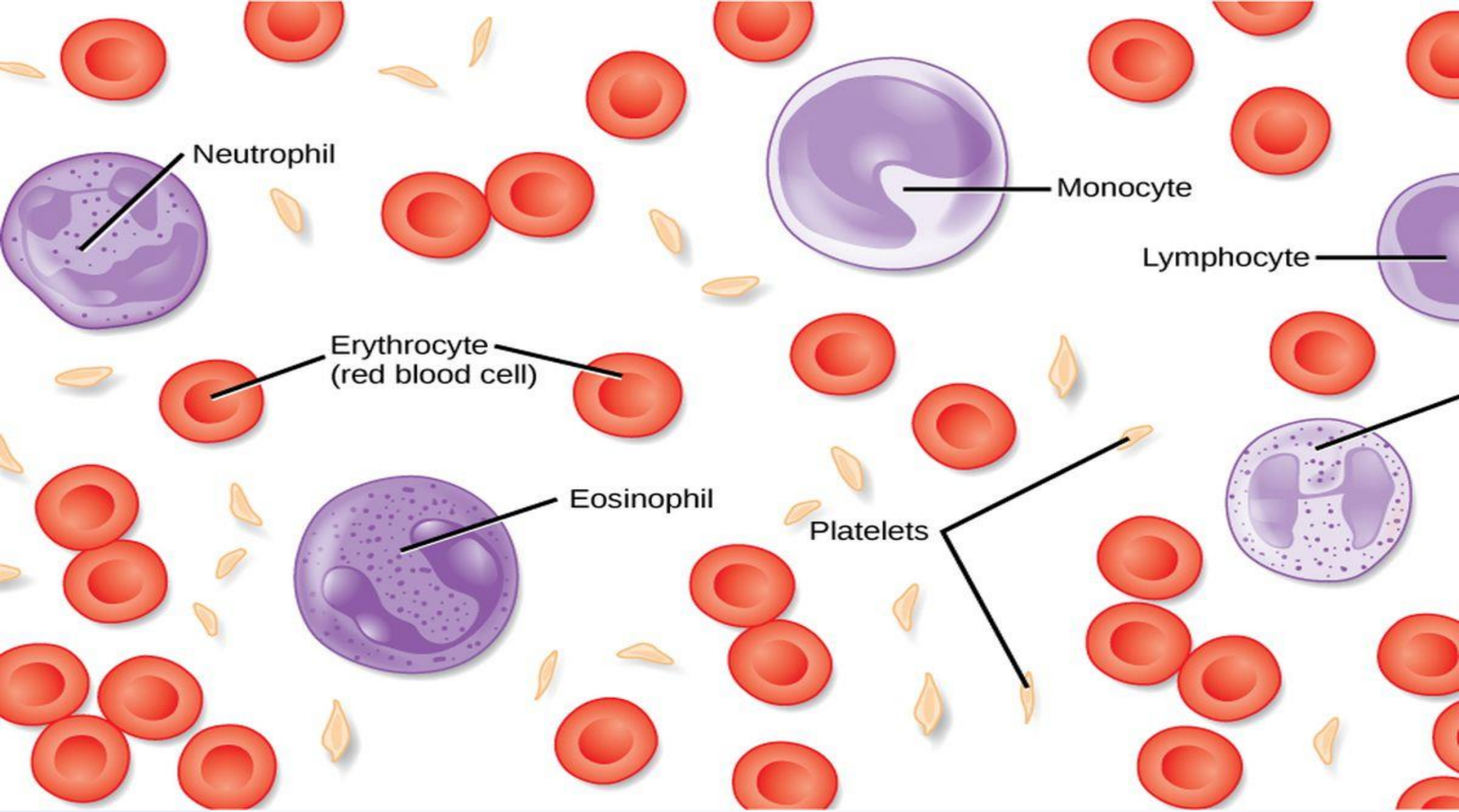


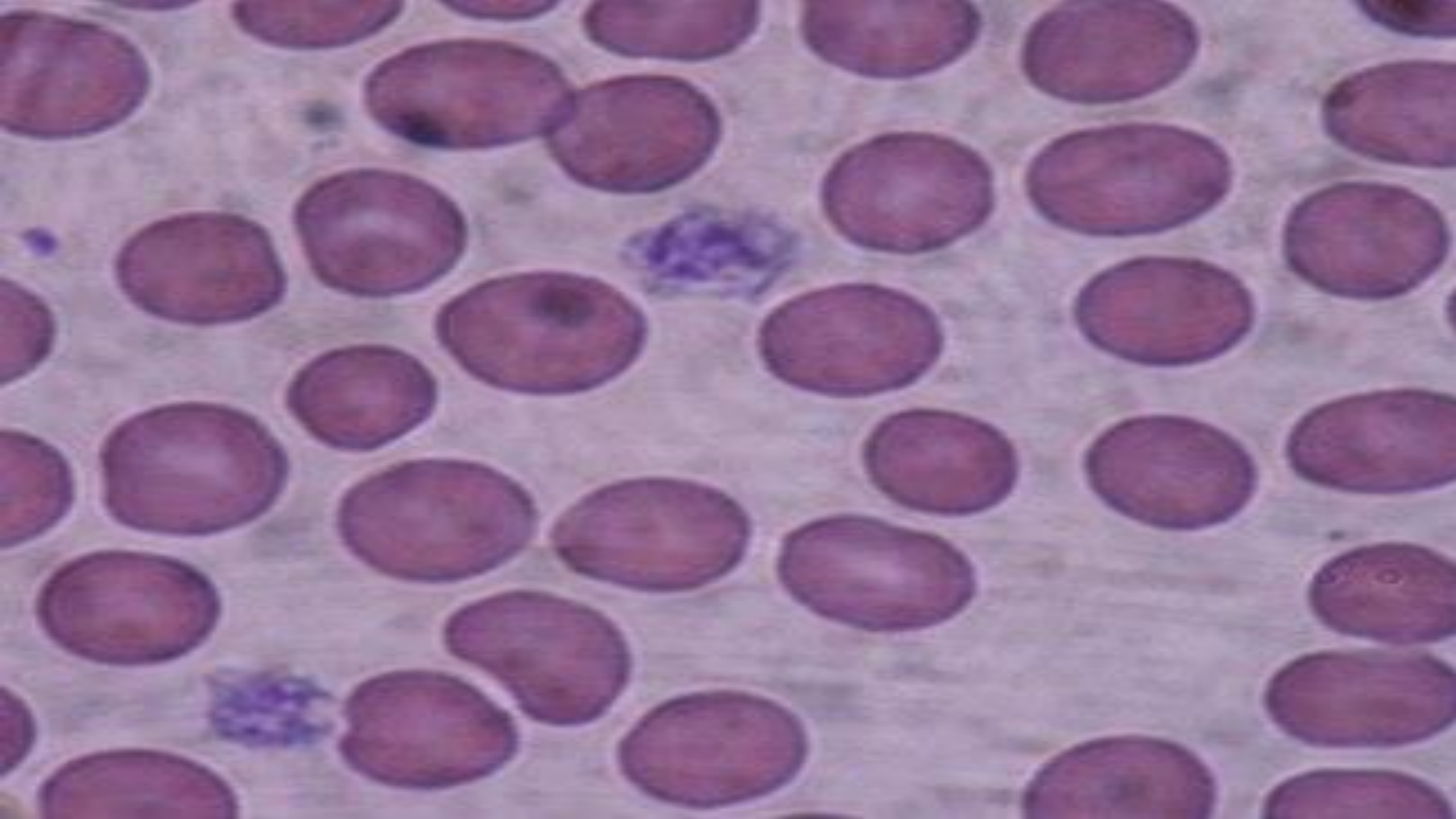
150–450 ribu

per µL darah

Kisaran rujukan umum, dapat berbeda antar-laboratorium.

Betts dkk., 2022; Keohane dkk., 2024


Neutrophil
Monocyte
Lymphocyte
Erythrocyte
(red blood cell)
Eosinophil
Platelets



Hemostasis: bagaimana darah berhenti?

Ketika pembuluh darah cedera, tubuh menjalankan respons bertahap.

1

VASOKONSTRIKSI

Pembuluh darah menyempit.

2

SUMBAT TROMBOSIT

Trombosit melekat dan berkumpul.

3

KOAGULASI

Fibrin memperkuat sumbat.

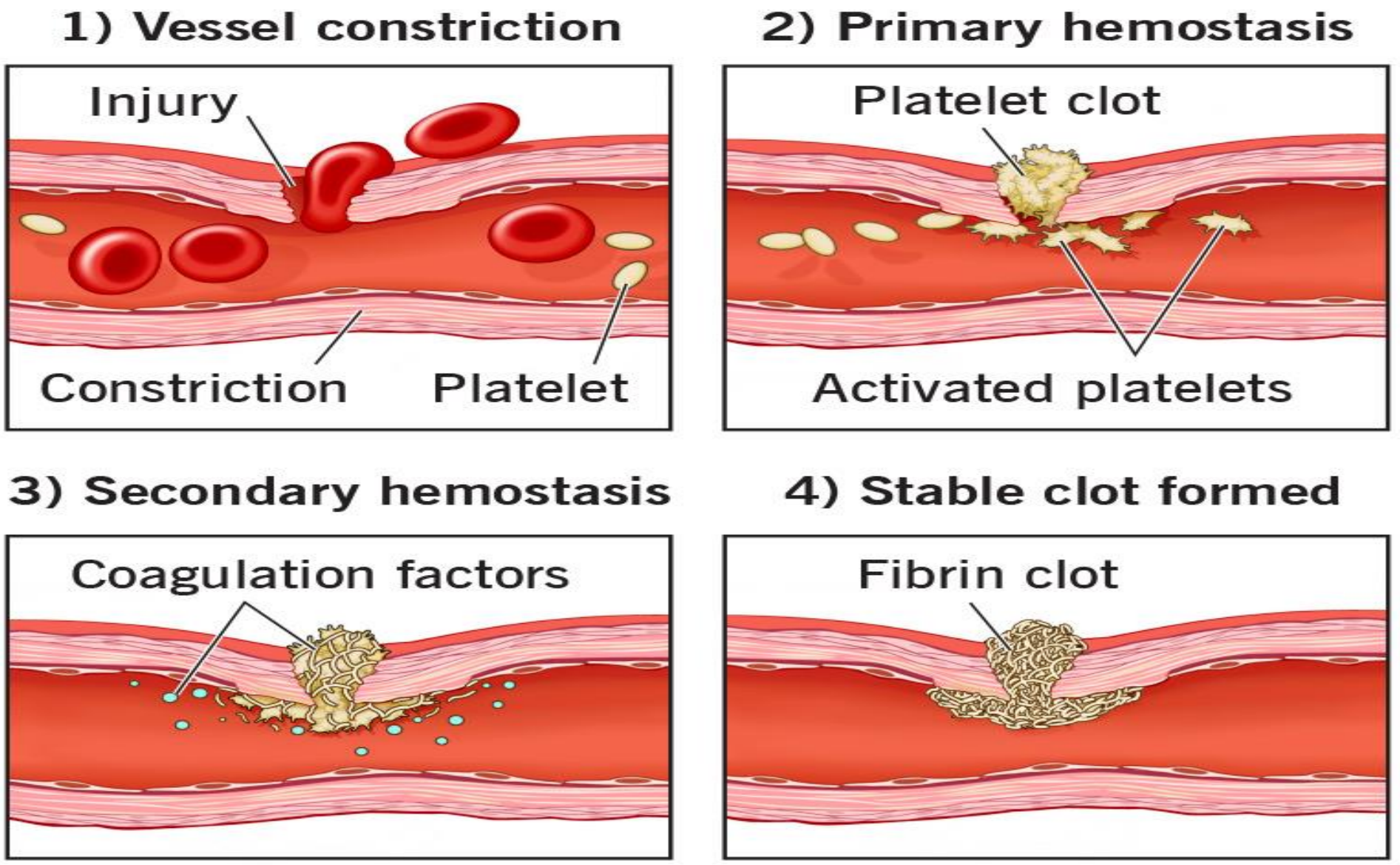
4

RESOLUSI

Bekuan diurai saat jaringan pulih.

Keohane dkk., 2024; Betts dkk., 2022

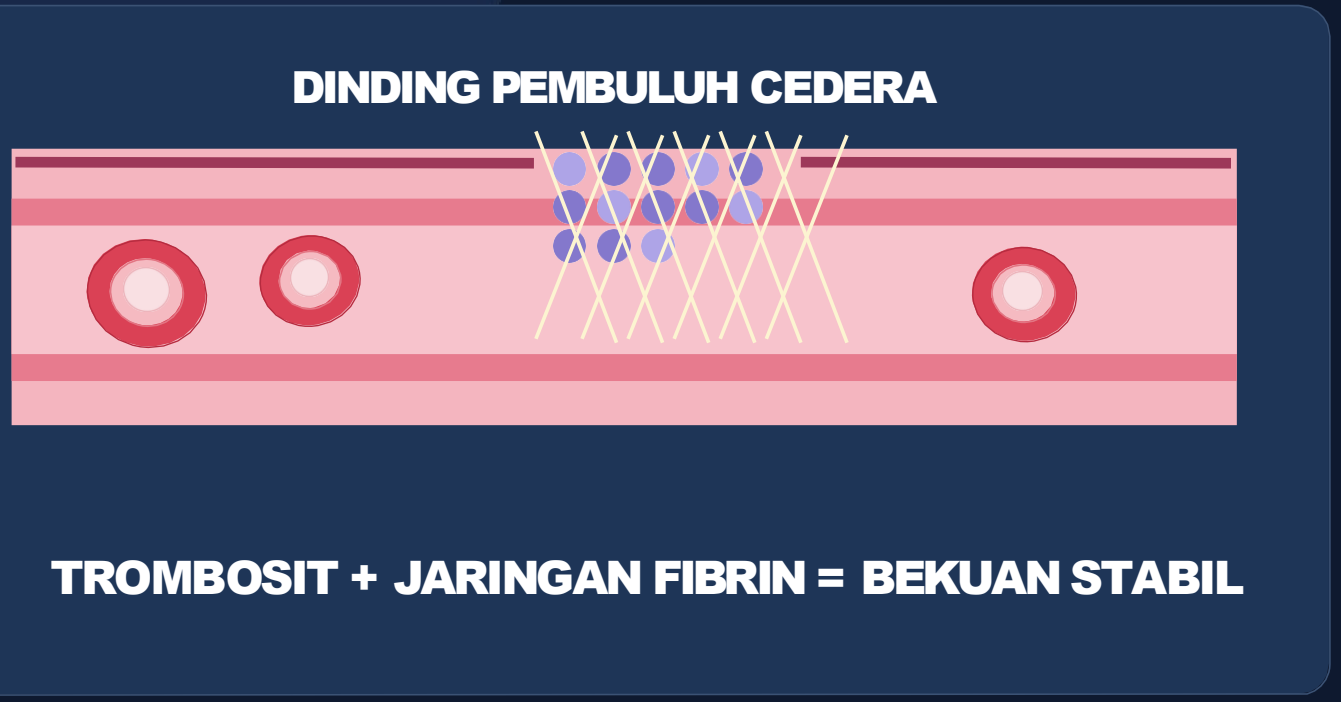
Hemostasis



Cleveland Clinic ©2024

Koagulasi: fibrin memperkuat bekuan

Sumbat trombosit terbentuk lebih dulu, kemudian jalinan fibrin menstabilkan bekuan.



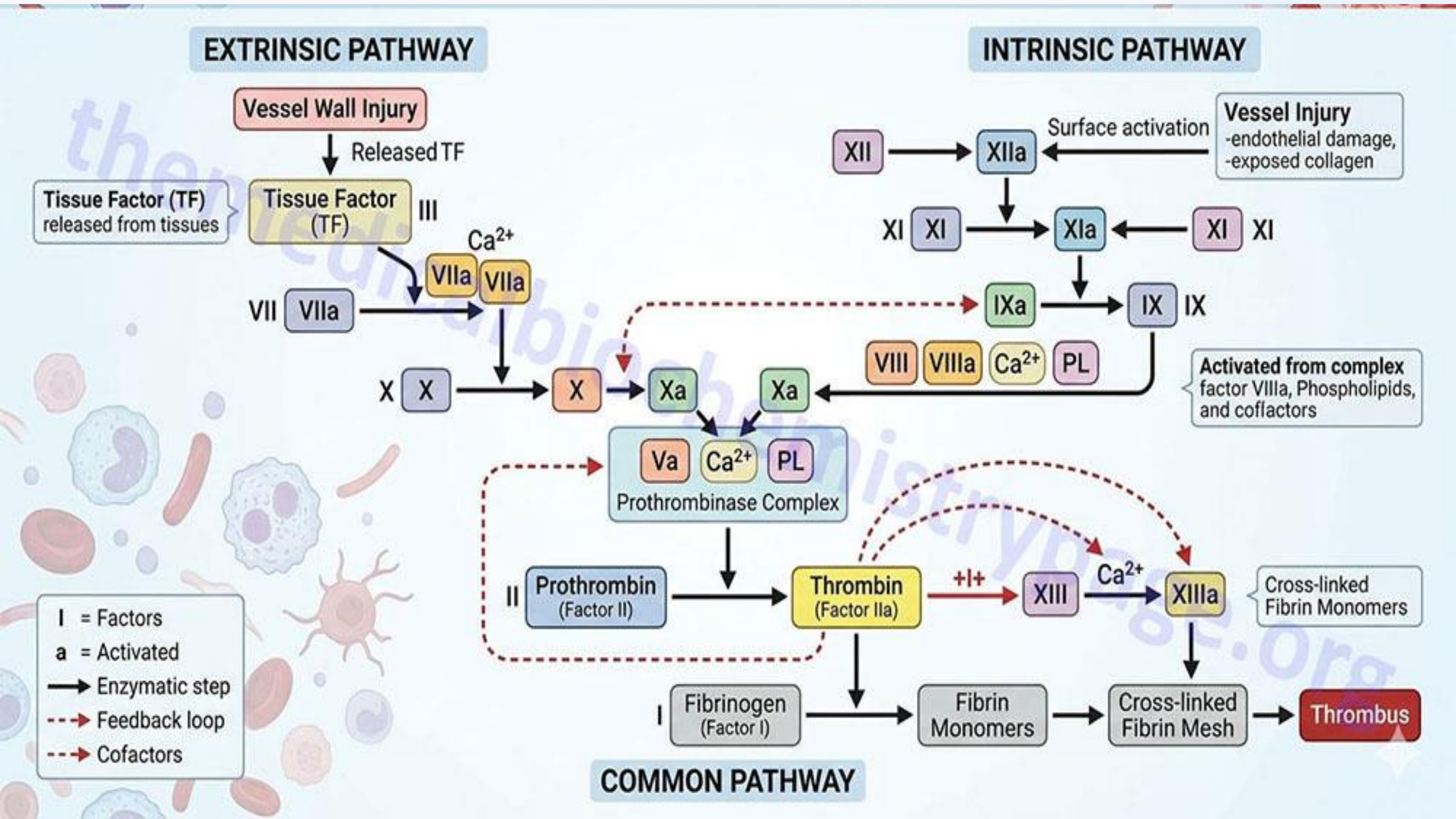
01 Kaskade dimulai
Paparan faktor jaringan memicu koagulasi.

02 Trombin terbentuk
Protrombin diubah menjadi trombin.

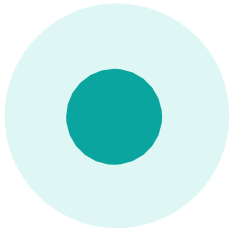
03 Fibrin terbentuk
Trombin mengubah fibrinogen menjadi fibrin.

Fibrinolisis membantu menguraikan bekuan setelah jaringan pulih.

Keohane dkk., 2024; Betts dkk., 2022

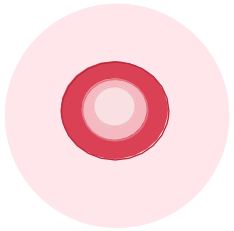

EXTRINSIC PATHWAY
Vessel Wall Injury
Released TF
Tissue Factor (TF) released from tissues
Tissue Factor (TF)
III
Ca2+
VIIa
VIIa
VII
VIIa
X
X
X
Xa
Xa
INTRINSIC PATHWAY
XII
XIIa
Surface activation
Vessel Injury
-endothelial damage,
-exposed collagen
XI
XI
XIa
XI
XI
IXa
IX
IX
VIII
VIIIa
Ca2+
PL
Activated from complex
factor VIIIa, Phospholipids, and cofactors
Va
Ca2+
PL
Prothrombinase Complex
II
Prothrombin
(Factor II)
Thrombin
(Factor IIa)
+I+
XIII
Ca2+
XIIIa
Cross-linked
Fibrin Monomers
I = Factors
a = Activated
Enzymatic step
Feedback loop
Cofactors
I
Fibrinogen
(Factor I)
Fibrin
Monomers
Cross-linked
Fibrin Mesh
Thrombus
COMMON PATHWAY

Sirkulasi darah & pertukaran gas



PARU-PARU

Darah memperoleh O_2 dan melepas CO_2.



JANTUNG

Memompa darah melalui sirkulasi paru dan sistemik.



JARINGAN

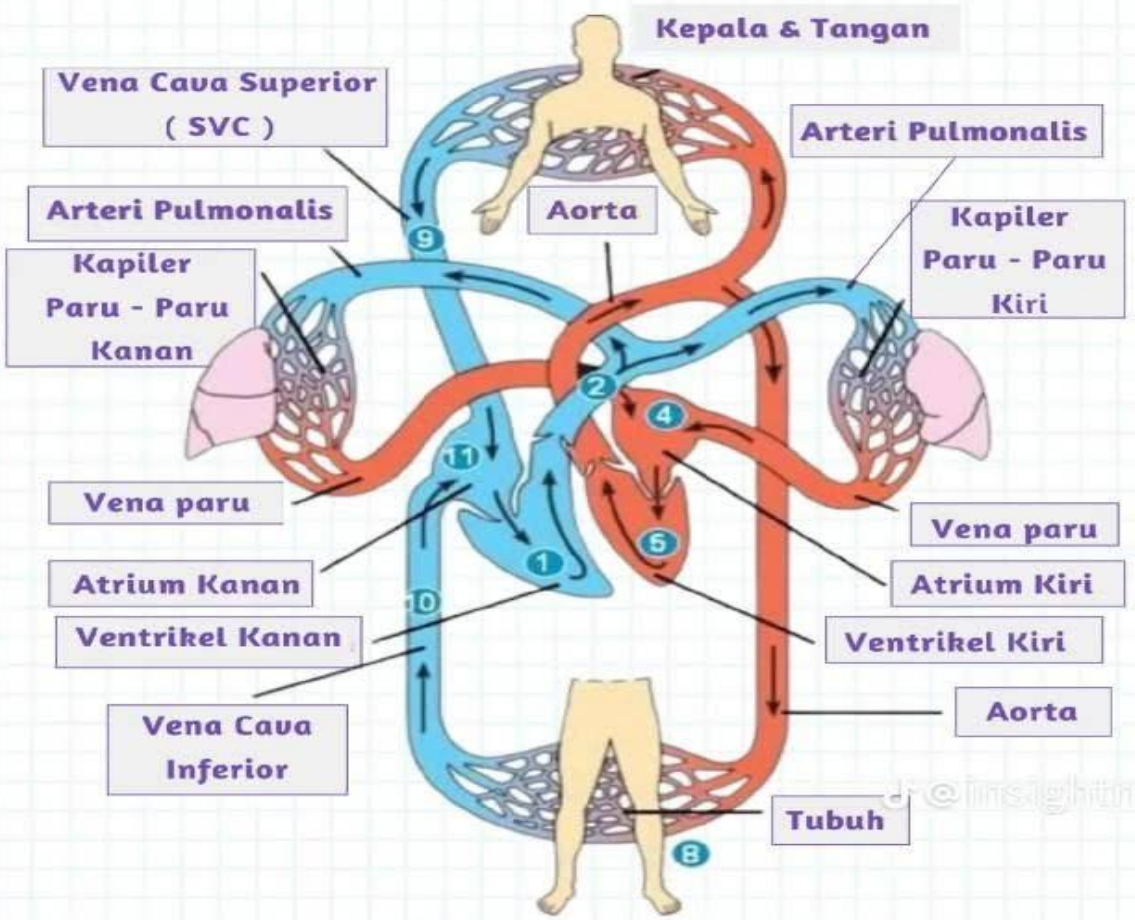
O_2 dan nutrisi disalurkan ke sel tubuh.

SISTEM KARDIOVASKULAR + SISTEM HEMATOLOGI = TRANSPORT TERINTEGRASI

Hall & Hall, 2025; Betts dkk., 2022

Sistem Kardiovaskuler

Rute Sirkulasi Peredaran Darah

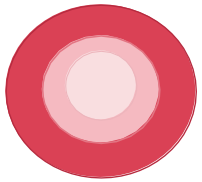


Merah : Mengandung O2

Biru : Mengandung CO2

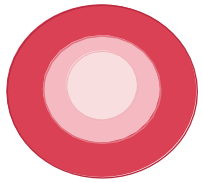
Golongan darah ABO dan Rhesus

Golongan darah ditentukan oleh antigen pada permukaan eritrosit.



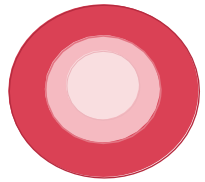
A

Antigen A



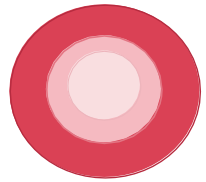
B

Antigen B



AB

Antigen A + B

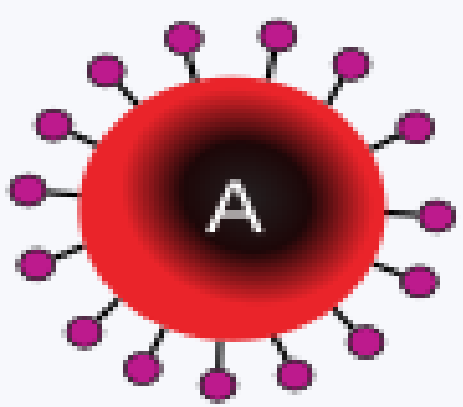
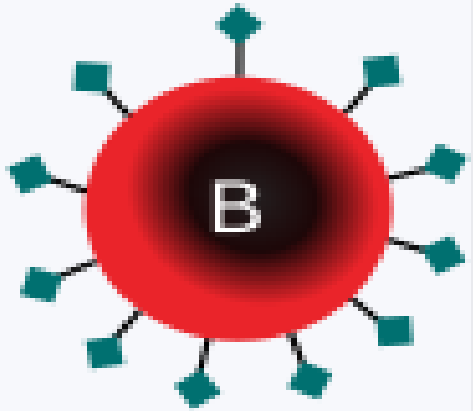
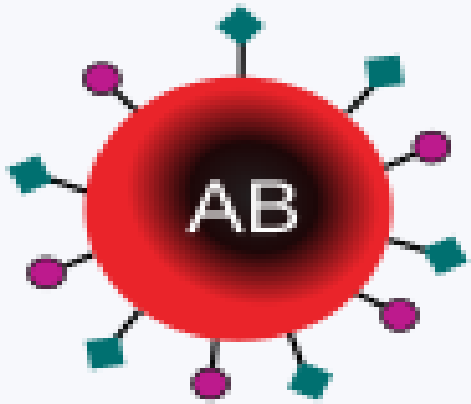
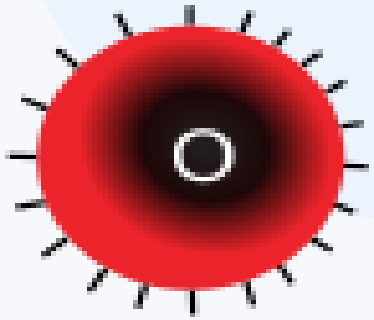
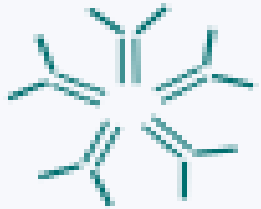
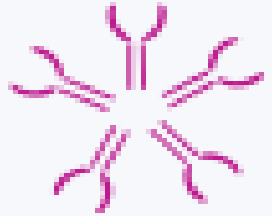


O

Tanpa antigen A / B

Rh positif (+): antigen D ada | Rh negatif (−): antigen D tidak ada

Hoffbrand dkk., 2024; Hall & Hall, 2025

	Group A	Group B	Group AB	Group O
Red blood cell type				
Antibodies in plasma	 Anti-B	 Anti-A	None	 Anti-A and Anti-B
Antigens in red blood cell	 A antigen	 B antigen	 A and B antigens	None

Apa yang dinilai pada hitung darah lengkap?

Hb

Kapasitas pengangkutan O_2

ERITROSIT

Jumlah dan karakteristik sel darah merah

LEUKOSIT

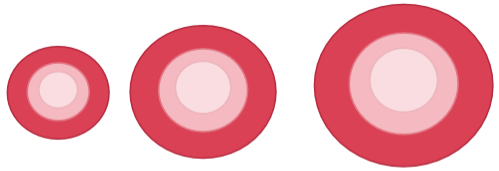
Jumlah sel pertahanan tubuh

TROMBOSIT

Jumlah komponen hemostasis

Indeks eritrosit: membaca pola anemia

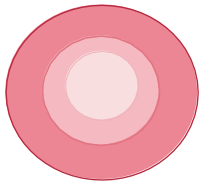
Setelah melihat Hb, periksa ukuran eritrosit, kadar Hb sel, dan variasi ukurannya.



MCV

Ukuran rata-rata eritrosit

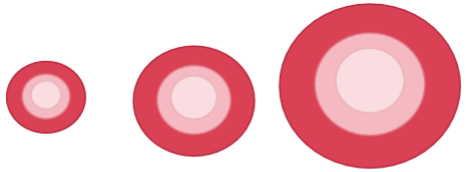
<80 fL mikro | 80-100 normal
>100 fL makro



MCHC

Konsentrasi Hb dalam eritrosit

Nilai rendah mendukung pola hipokromik



RDW

Variasi ukuran eritrosit

RDW meningkat: variasi ukuran makin besar

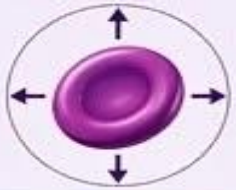
RETIKULOSIT: membantu menilai respons produksi eritrosit oleh sumsum tulang.

Keohane dkk., 2024; Hoffbrand dkk., 2024

MCV vs MCH vs MCHC

Understanding Red Blood Cell Indices in CBC

MCV
(Mean Corpuscular Volume)



What it measures:
The average size (volume) of a red blood cell (RBC)

- **Unit:** femtoliters (fL)
- **Normal range:** 80 – 100 fL
- **Clinical use:**
  - Low MCV → microcytic anemia (e.g., iron deficiency)
  - High MCV → macrocytic anemia (e.g., vitamin B12 deficiency)

MCH
(Mean Corpuscular Hemoglobin)



What it measures:
The average amount of hemoglobin per RBC

- **Unit:** picograms (pg)
- **Normal range:** 27 – 33 pg
- **Clinical use:**
  - Low in hypochromic anemia
  - Often parallels MCV (small cells → less Hb)

MCHC
(Mean Corpuscular Hemoglobin Concentration)



What it measures:
The concentration of hemoglobin inside RBCs

- **Unit:** g/dL
- **Normal range:** 32 – 36 g/dL
- **Clinical use:**
  - Low MCHC → hypochromia (pale RBCs)
  - High MCHC → seen in conditions like spherocytosis (rare)

KEY DIFFERENCES

Parameter	Measures	Focus	Clinical Meaning
MCV	Size of RBC	Cell volume	Type of anemia (micro/macrocytic)
MCH	Amount of Hb per cell	Hb weight	Hb content per RBC
MCHC	Hb concentration	Hb density	Color/intensity of RBC

Easy Way to Remember

MCV = Size | **MCH** = Amount of Hb | **MCHC** = Concentration of Hb

MedicalLabScientist
SCIENCE • ACCURACY • HEALTH

1. HEMATOLOGY FORMULAS

1 MEAN CORPUSCULAR VOLUME (MCV)

Purpose: Determines average size of red blood cells.

$$\text{MCV (fL)} = \frac{\text{Hematocrit (\%)} \times 10}{\text{RBC count (million/µL)}}$$

TYPICAL ADULT RANGE:
80 – 100 fL
Ranges may vary by laboratory.

2 MEAN CORPUSCULAR HEMOGLOBIN (MCH)

Purpose: Determines average amount of hemoglobin per red blood cell.

$$\text{MCH (pg)} = \frac{\text{Hemoglobin (g/dL)} \times 10}{\text{RBC count (million/µL)}}$$

TYPICAL ADULT RANGE:
27 – 33 pg
Ranges may vary by laboratory.

3 MEAN CORPUSCULAR HEMOGLOBIN CONCENTRATION (MCHC)

Purpose: Measures average hemoglobin concentration within RBCs.

$$\text{MCHC (g/dL)} = \frac{\text{Hemoglobin (g/dL)} \times 100}{\text{Hematocrit (\%)}}$$

TYPICAL ADULT RANGE:
32 – 36 g/dL
Ranges may vary by laboratory.

4 RED CELL DISTRIBUTION WIDTH (RDW-CV)

Purpose: Measures variation in RBC size (anisocytosis).

$$\text{RDW-CV (\%)} = \frac{\text{SD of RBC volume}}{\text{MCV}} \times 100$$

TYPICAL ADULT RANGE:
11.5 – 14.5%
Ranges may vary by laboratory.

5 CORRECTED WBC COUNT

Purpose: Adjusts WBC count when nucleated RBCs are present.

$$\text{Corrected WBC} = \frac{\text{Uncorrected WBC} \times 100}{100 + \text{NRBCs per 100 WBCs}}$$

6 ABSOLUTE NEUTROPHIL COUNT (ANC)

Purpose: Helps assess neutropenia and infection risk.

$$\text{ANC} = \text{WBC} \times \frac{(\text{\% Neutrophils} + \text{\% Bands})}{100}$$

TYPICAL ADULT RANGE:
1,500 – 8,000 cells/µL
Ranges may vary by laboratory.

7 CORRECTED RETICULOCYTE COUNT

Purpose: Evaluates bone marrow response by adjusting for hematocrit.

$$\text{Corrected Retic (\%)} = \text{Retic \%} \times \frac{\text{Patient Hct}}{\text{Normal Hct}}$$

8 RETICULOCYTE PRODUCTION INDEX (RPI)

Purpose: Assesses adequacy of erythropoiesis.

$$\text{RPI} = \frac{\text{Corrected Retic \%}}{\text{Maturation Correction Factor}}$$

MATURATION CORRECTION FACTOR

Hct ≥ 45%	= 1.0
Hct 35 – 44%	= 1.5
Hct 25 – 34%	= 2.0
Hct < 25%	= 2.5

Note: These are typical adult reference ranges.
Results should be interpreted in the context of clinical findings and laboratory specific reference intervals.

Ambang hemoglobin untuk anemia

PEDOMAN WHO 2024

PRIA DEWASA	WANITA TIDAK HAMIL	HAMIL TRIMESTER II
< 13,0 g/dL	< 12,0 g/dL	< 10,5 g/dL

Kehamilan trimester I dan III: anemia bila Hb < 11,0 g/dL.

Studi kasus: membaca hasil laboratorium

KASUS ILUSTRATIF

Perempuan, 24 tahun

Keluhan: mudah lelah dan pusing.

Hb	9,5 g/dL
MCV	72 fL
Ferritin	8 ng/mL

Pola mengarah ke anemia mikrositik. Evaluasi penyebab.

WHO, 2024; Keohane dkk., 2024 | Kasus ilustrasi

Contoh gangguan sistem hematologi

ANEMIA

Penurunan kadar hemoglobin sesuai kriteria.

LEUKEMIA

Keganasan yang melibatkan sel pembentuk darah.

TROMBOSITOPENIA

Jumlah trombosit menurun dan bisa meningkatkan risiko perdarahan.

GANGGUAN KOAGULASI

Kelainan proses pembentukan atau pengaturan bekuan.

Hoffbrand dkk., 2024; Keohane dkk., 2024

Empat konsep yang perlu diingat

01

DARAH

Media transport, pertahanan, dan regulasi.

02

SUMSUM TULANG

Tempat utama pembentukan sel darah.

03

SEL DARAH

Eritrosit, leukosit, trombosit memiliki fungsi berbeda.

04

HEMOSTASIS

Trombosit dan koagulasi mengendalikan perdarahan.

Betts dkk., 2022; Hoffbrand dkk., 2024

TERIMA KASIH

Diskusi dan tanya jawab

dr. Juwita Ninda, M.Kes

INSTITUT KESEHATAN RAJAWALI

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